Programming for Problem Solving Using Python
Unit-1

Session-7

Numeric and Boolean Types

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2025

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Numeric and Boolean Types

Immutable Types- Numbers

- Numbers in Python refer to the numeric data types in Python programming. Python supports three kinds of numeric data types: int, float, and complex.
- Integers and floating points are separated by the presence or absence of a decimal point. For instance, 5 is an integer and 5.42 is a floating-point number.

Immutable Types- Numbers

- Python int is the whole number, including negative numbers but not fractions. In Python, there is no limit to how long an integer value can be.
- There are three built-in number types available in Python: integers (int), floating point numbers (float) and complex numbers.

Immutable Types- Numbers

Example:

```
num1 = 5
print(num1, 'is of type', type(num1))

num2 = 5.42
print(num2, 'is of type', type(num2))

num3 = 8+2j
print(num3, 'is of type', type(num3))
```

```
5 is of type <class 'int'>
5.42 is of type <class 'float'>
(8+2j) is of type <class 'complex'>
```

Numbers- Integers

Python - Integer Numbers

- In Python, any number without the provision to store a fractional part is an integer. (Note that if the fractional part in a number is 0, it doesn't mean that it is an integer.
- For example a number 10.0 is not an integer, it is a float with 0 fractional part whose numeric value is 10.) An integer can be zero, positive or a negative whole number. For example, 1234, 0, -55 all represent to integers in Python.

Numbers- Integers

- There are three ways to form an integer object. With (a) literal representation, (b) any expression evaluating to an integer, and (c) using int() function.
- Literal is a notation used to represent a constant directly in the source code. For example -

Understanding Boolean Type

- Boolean is a **data type** with only two possible values: `True` or `False`.
- Used to represent **truth values** in logic and decision making.
- Booleans often result from **comparisons.**

```python
x = 10
y = 5
print(x > y)  # Output: True
print(x == y) # Output: False
```

```
True
False
```

Boolean Logic and Control Flow

- Booleans are **essential for decision-making** in Python
- Common logical operators:
  - `and`: True if both are True
  - `or`: True if at least one is True
  - `not`: Inverts the value

Test Your Knowledge- Fill in the Blanks

1. Immutable types in Python are objects whose state or value cannot be _____after they are created.
2. Examples of immutable types in Python include _____, _____, and _____.
3. Strings in Python are _____.
4. Once a tuple is created in Python, its elements cannot be _____.
5. Immutable types provide _____and _____ in terms of memory management.

Test Your Knowledge- MCQ

3. What is the output of the following Python code snippet?

```
my_tuple = (1, 2, 3)
print(my_tuple[1])
```

   a. 1
   b. 2
   c. 3
   d. Error: Tuples are immutable

4. How can you concatenate two strings in Python?

   a. Using the concat() method
   b. Using the join() method
   c. Using the & operator
   d. Using the + operator

Test Your Knowledge- MCQ

5. What is the output of the following Python code snippet?

```
string = "Hello, World!"
print(string[3:7])
```

a. "Hello"
b. "lo, "
c. "lo, W"
d. "lo, World"

Test Your Knowledge- MCQ (Solutions)

1. d. Tuples are ordered collections of elements
2. b. Deletion of an element
3. b. 2
4. d. Using the + operator
5. b. “lo, ”

Test Your Knowledge- Interview Questions

1. What is an immutable type in Python?
2. Why are immutable types important in programming?
3. Give examples of immutable types in Python.
4. What is the difference between mutable and immutable types?
5. Explain why strings are immutable in Python.
6. Can you change the value of a string in Python after it has been created? Why or why not?
7. Discuss the immutability of tuples in Python and how it differs from lists.

Test Your Knowledge- Interview Questions (Solutions)

1. An immutable type in Python is an object whose state or value cannot be modified after it is created.
2. Immutable types are important because they prevent unexpected side effects, are hashable, and are safe for concurrent programming.
3. Examples include integers (‘int‘), floating-point numbers (‘float‘), strings (‘str‘), tuples (‘tuple‘), frozen sets (‘frozenset‘), and booleans (‘bool‘).

Test Your Knowledge- Interview Questions (Solutions)

4. Mutable types can be changed after creation (e.g., lists, dictionaries), while immutable types cannot be changed once created (e.g., strings, tuples).
5. Strings are immutable for performance optimization, safety, and simplicity. Immutable strings prevent accidental changes and allow optimizations like interning.
6. No, because strings are immutable. Any operation that modifies a string creates a new string object.

Test Your Knowledge- Interview Questions (Solutions)

7. Tuples are immutable and their elements cannot be changed once created, making them stable and hashable. Lists are mutable and can be modified after creation, providing flexibility but requiring careful management.

Thank You!

Any Questions?